Lec-11

22/05/26 7:06 PM

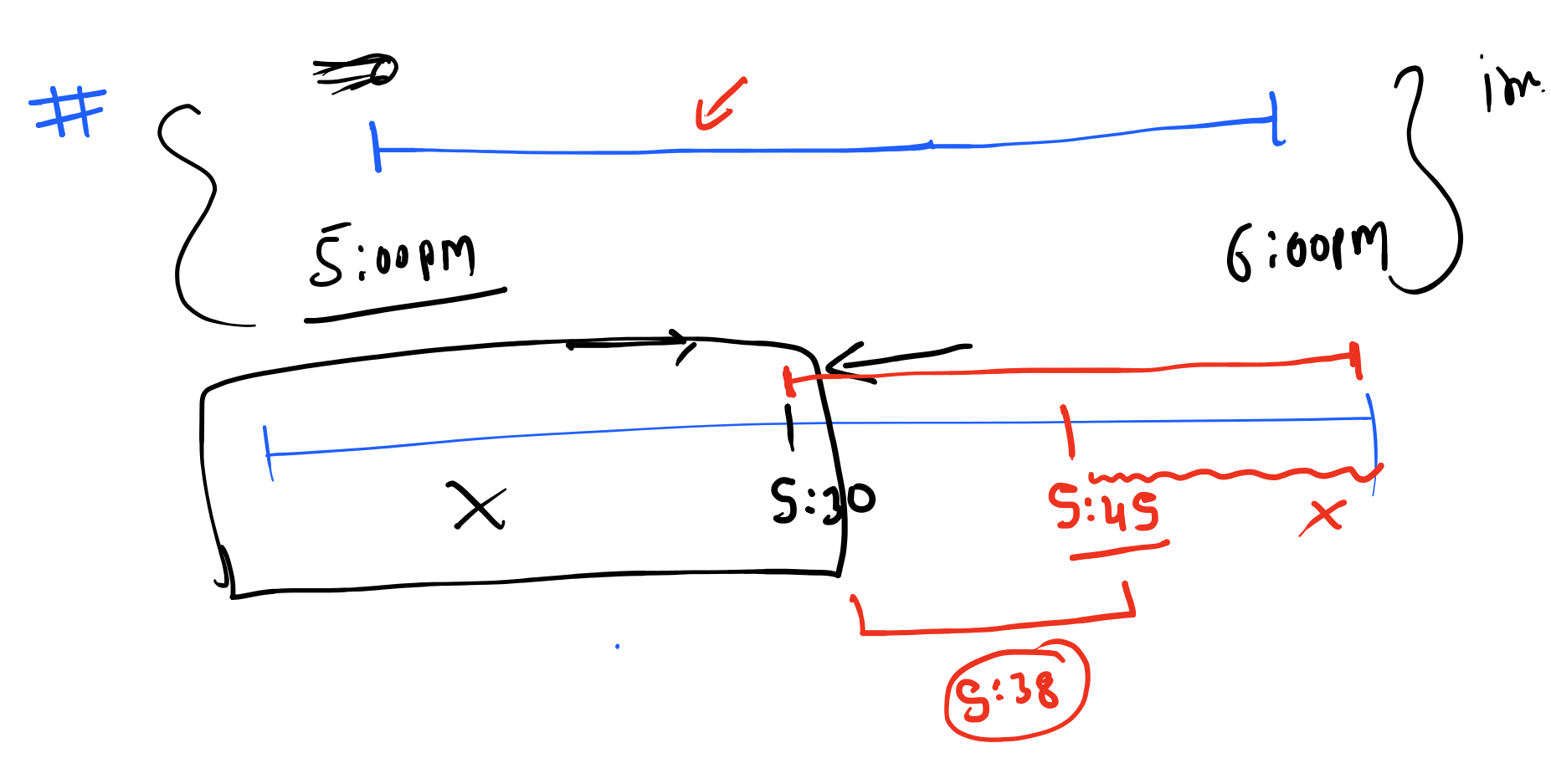
Binary-Search → condition

Linear Search

[2 3 6 9 13 15 18] ✓

[6 9 7 2 18 15 13] ✗

#



5:00PM
6:00PM
✗
5:30
5:45
✗
5:38

im

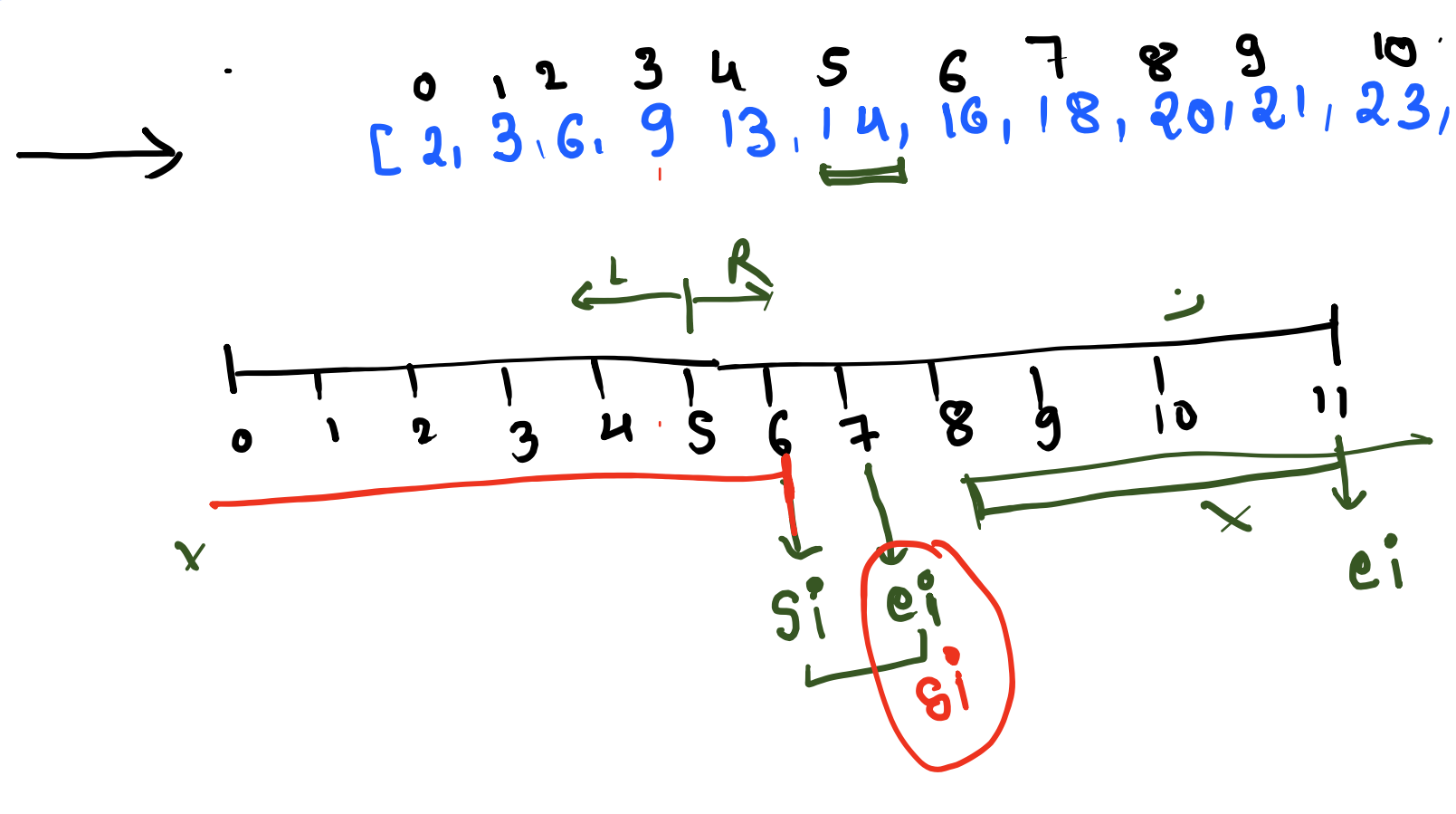
Big-code

ae-AT ②

0 1 2 3 4 5 6 7 8 9 10 11

[2, 3, 6, 9, 13, 14, 16, 18, 20, 21, 23, 25]

item = 18



L R
0 1 2 3 4 5 6 7 8 9 10 11
si
ei
si

$\frac{11+0}{2} = 5$

$\frac{11+6}{2} = 8$

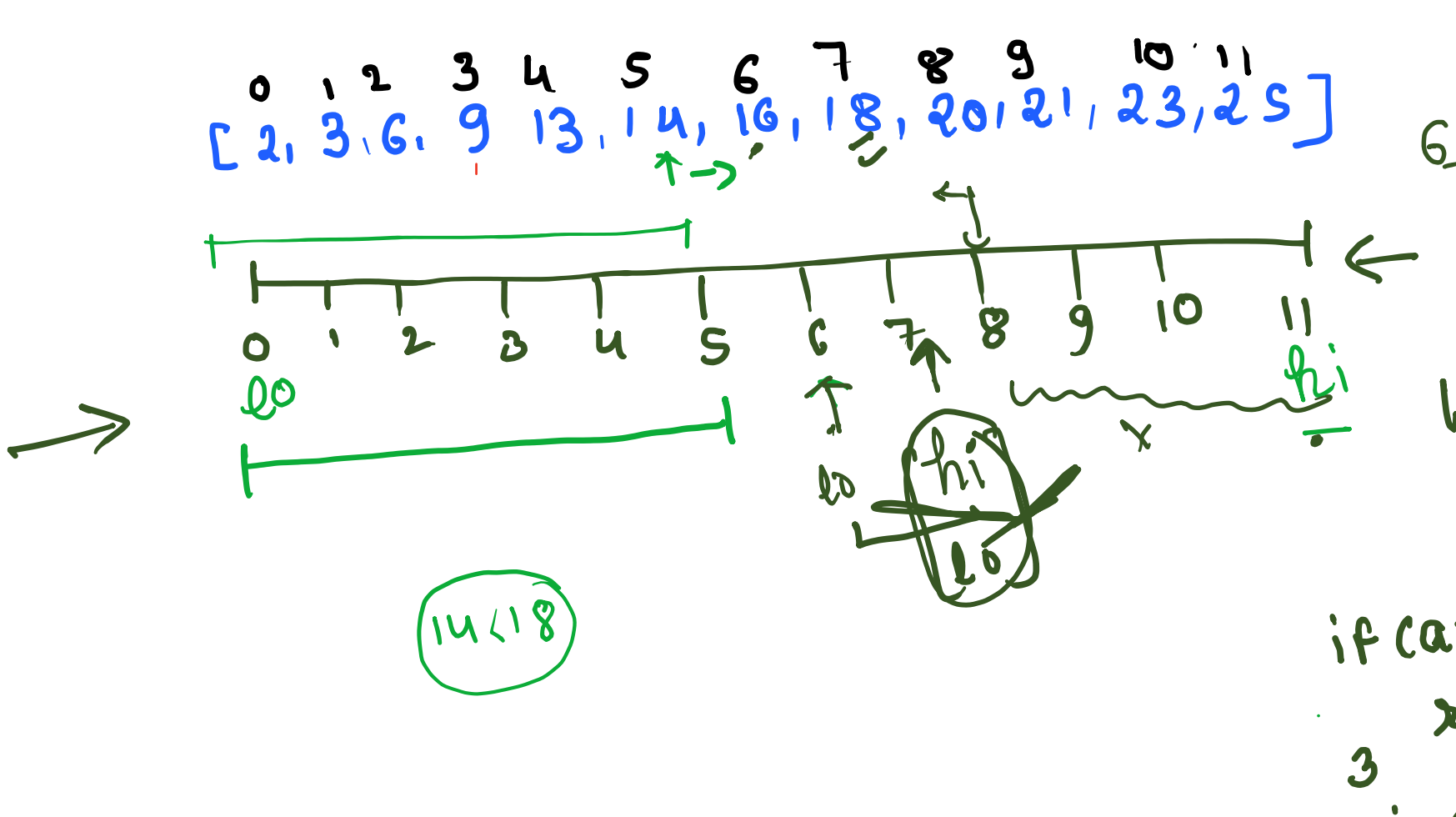
$\frac{6+7}{2} = 6$

0 1 2 3 4 5 6 7 8 9 10 11

[2, 3, 6, 9, 13, 14, 16, 18, 20, 21, 23, 25]

$\frac{6+11}{2} = 8$

$\frac{0+11}{2} = 5$ ✓



0 1 2 3 4 5 6 7 8 9 10 11
lo
lo
hi
lo
hi

14<18

```
while (lo<=hi) {
    mid = (lo+hi)/2
    if (arr[mid] == item) {
        ret mid
    }
    else if (arr[mid] > item) {
        hi = mid-1
    }
    else {
        lo = mid+1
    }
}
```

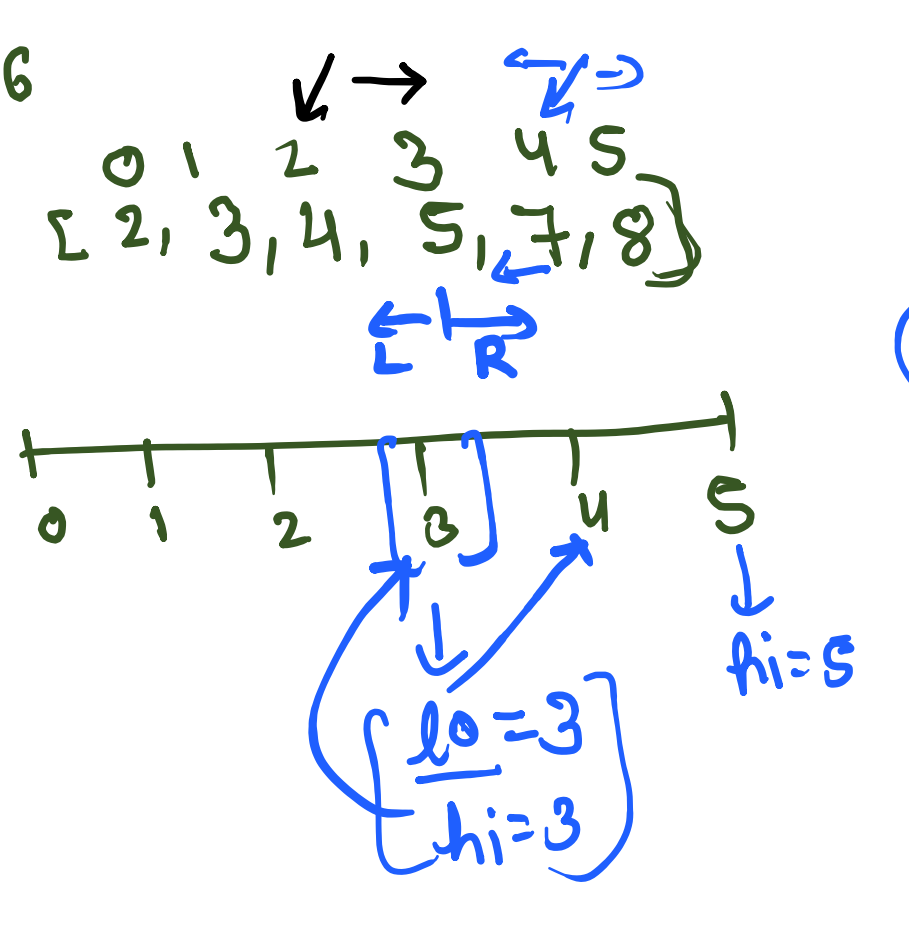
$\frac{0+5}{2} = 2$

```
while (lo<=hi) {
    mid = (lo+hi)/2
    if (arr[mid] == item) {
        ret mid
    }
    else if (arr[mid] > item) {
        hi = mid-1
    }
    else {
        lo = mid+1
    }
}
```

item=6

0 1 2 3 4 5

[2, 3, 4, 5, 7, 8]



L R
0 1 2 3 4 5
hi=5
lo=3
hi=3

④

$\frac{3+5}{2} = 4$

f→ $\boxed{x^k \le N}$

$x=1 \quad 1^3 \le 149$ ✓

$x=2 \quad 2^3 \le 149$ ✓

$x=3 \quad 3^3 \le 149$ ✓

$x=4 \quad 4^3 \le 149$ ✓

$x=5 \quad 5^3 \le 149$

$x=6 \quad 6^3 \le 149$ ✗

k=3

N=149

1 2 4 3 5

maxi

K ✓

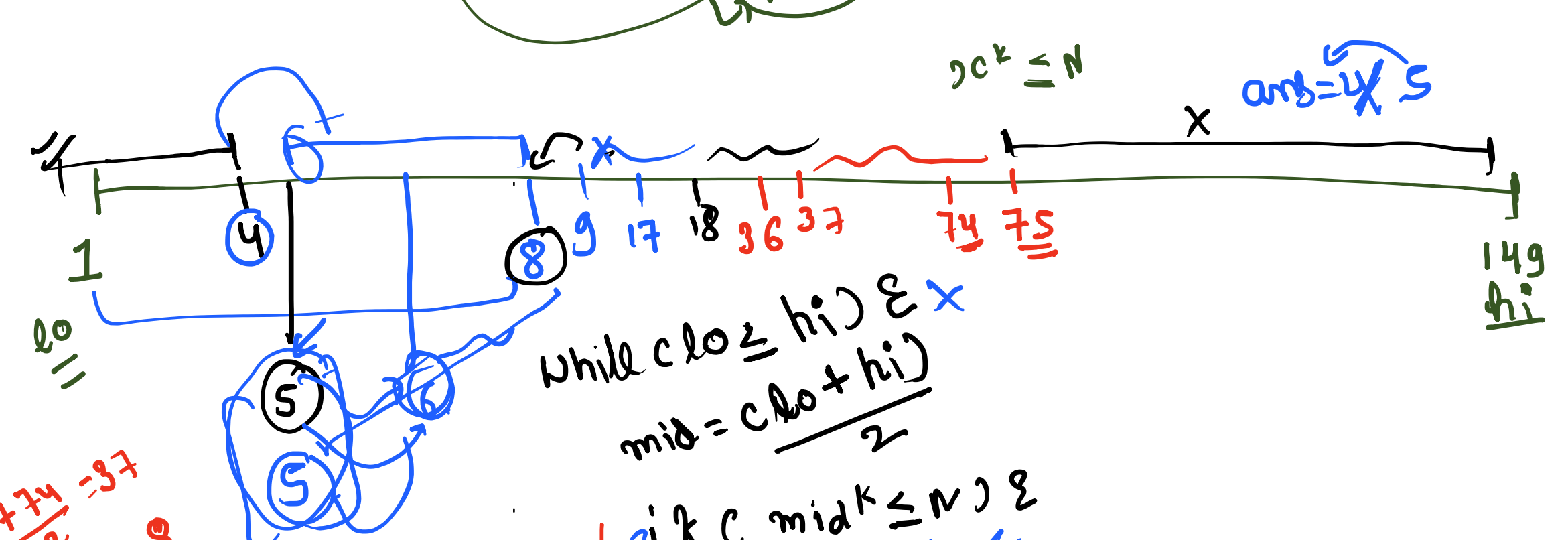
n ✓

maximum of x

109

① ✓

```
x = 1
while (x^k <= N) {
    x++
}
return x-1
```



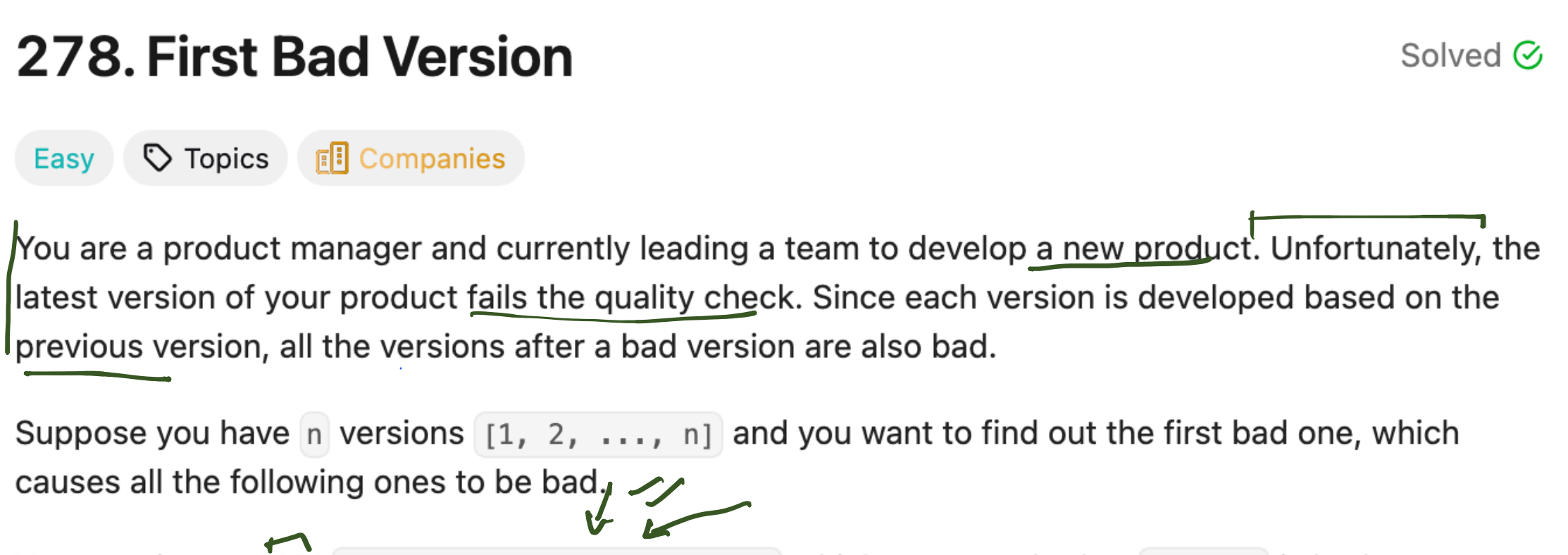
$x^k \le N$
✗
ans = 4 5
1
lo
4
5
5
6
8
9 17 18 36 37 74 75
149
hi

```
while (lo<=hi) {
    mid = (lo+hi)/2
    if (mid^k <= N) {
        ans = mid
        lo = mid+1
    }
    else
        hi = mid-1
}
```

$\frac{1+74}{2} = 37$

$\frac{1+36}{2} = 18$

$\frac{1+17}{2} = 9$



278. First Bad Version

Solved ✓

Easy · Topics · Companies

You are a product manager and currently leading a team to develop a new product. Unfortunately, the latest version of your product fails the quality check. Since each version is developed based on the previous version, all the versions after a bad version are also bad.

Suppose you have `n` versions `[1, 2, ..., n]` and you want to find out the first bad one, which causes all the following ones to be bad.

You are given an API `bool isBadVersion(version)` which returns whether `version` is bad. Implement a function to find the first bad version. You should minimize the number of calls to the API.

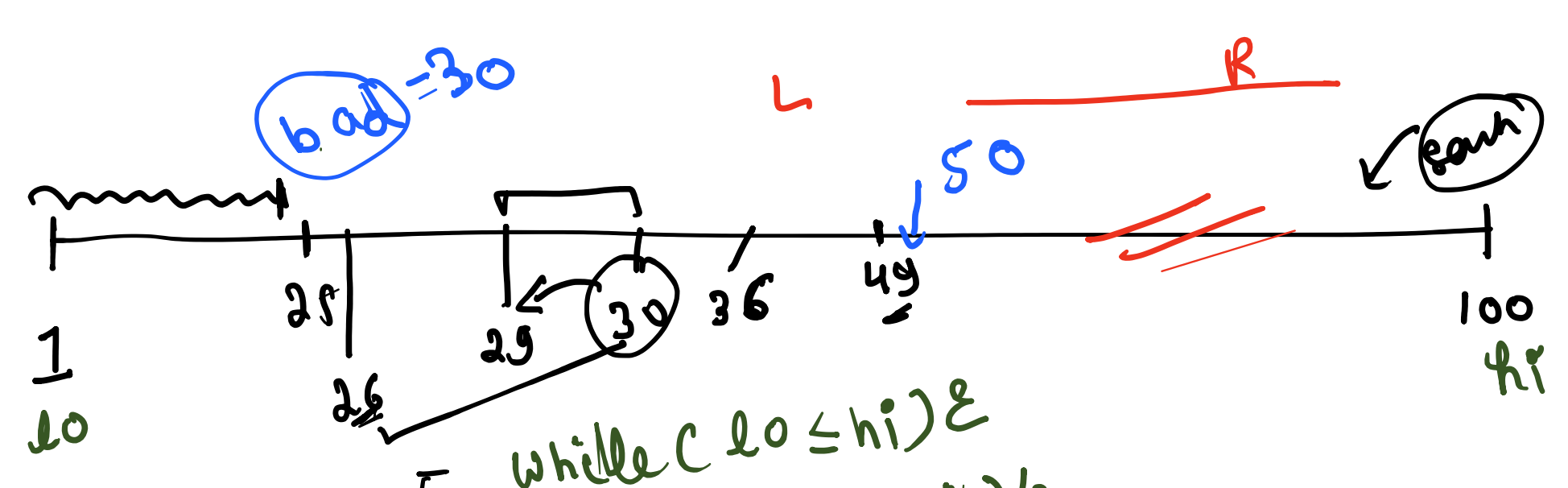
N=100

[1, 2, 3, 4, 5, 6, 7, 8, 9 ----- 100]

N=100 bad=30

```
for (i=1 i<=N i++) {
    if (isbadversion(i)) {
        return i
    }
}
```

bad=30



L R
1
lo
25
26
29
30
36
49
50
100
hi
bad

ans = ~~50~~ ~~37~~ 30 ~~31~~

```
while (lo<=hi) {
    mid = (lo+hi)/2
    if (isbadversion(mid)) {
        ans = mid
        hi = mid-1
    }
    else
        lo = mid+1
}
```

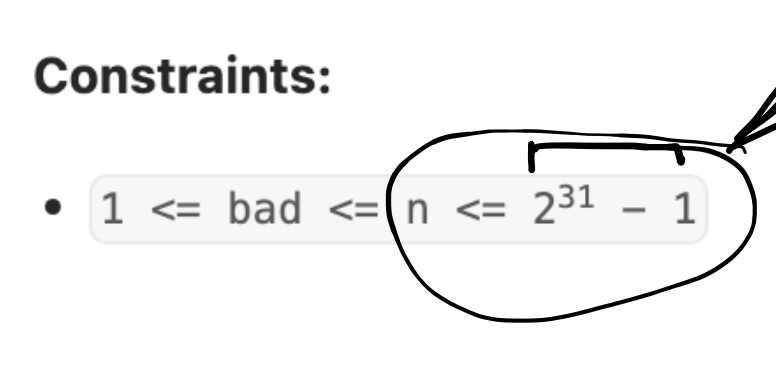
25

$\frac{26+49}{2} = 37$

$\frac{26+36}{2} = 31$

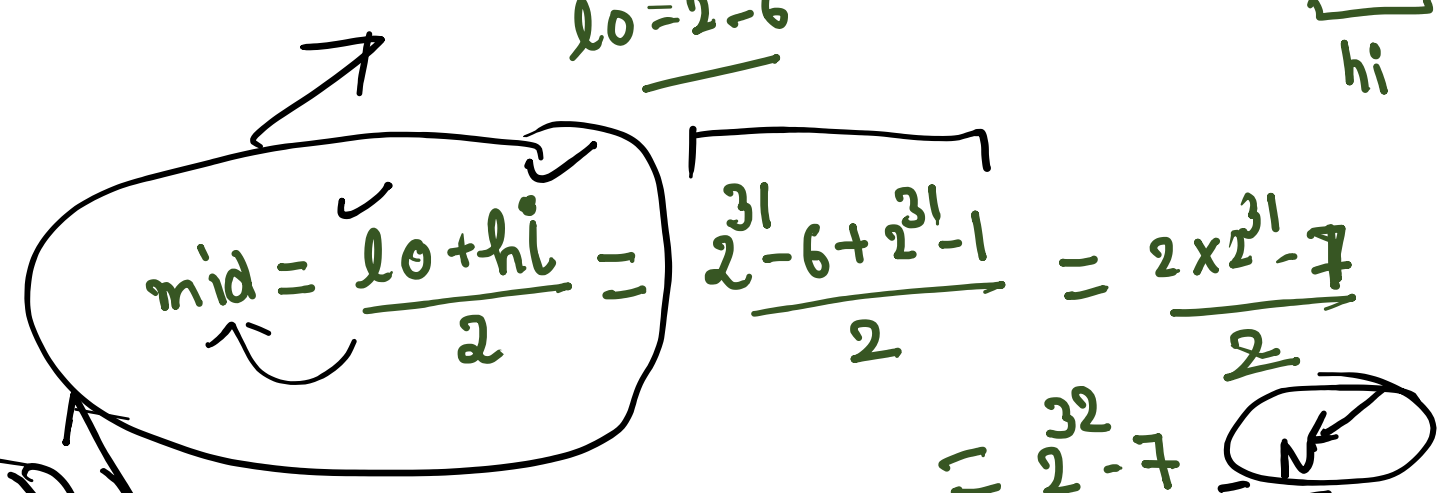
100

$bad = 2^{31}-3$ $\quad N = 2^{31}-1$ ✓



Constraints:

- `1 <= bad <= n <= 2^31 - 1`



Mid
1
$lo = 2^{31}-6$
$2^{31}-1$
hi

$$mid = \frac{lo+hi}{2} = \frac{2^{31}-6+2^{31}-1}{2} = \frac{2\times 2^{31}-7}{2} = \frac{2^{32}-7}{2}$$

$2^{31}-3$ mid

$$\frac{(2^{31}-1)-(2^{31}-6)}{2} + 2^{31}-6$$

$$\frac{-1+6}{2} + 2^{31}-6$$

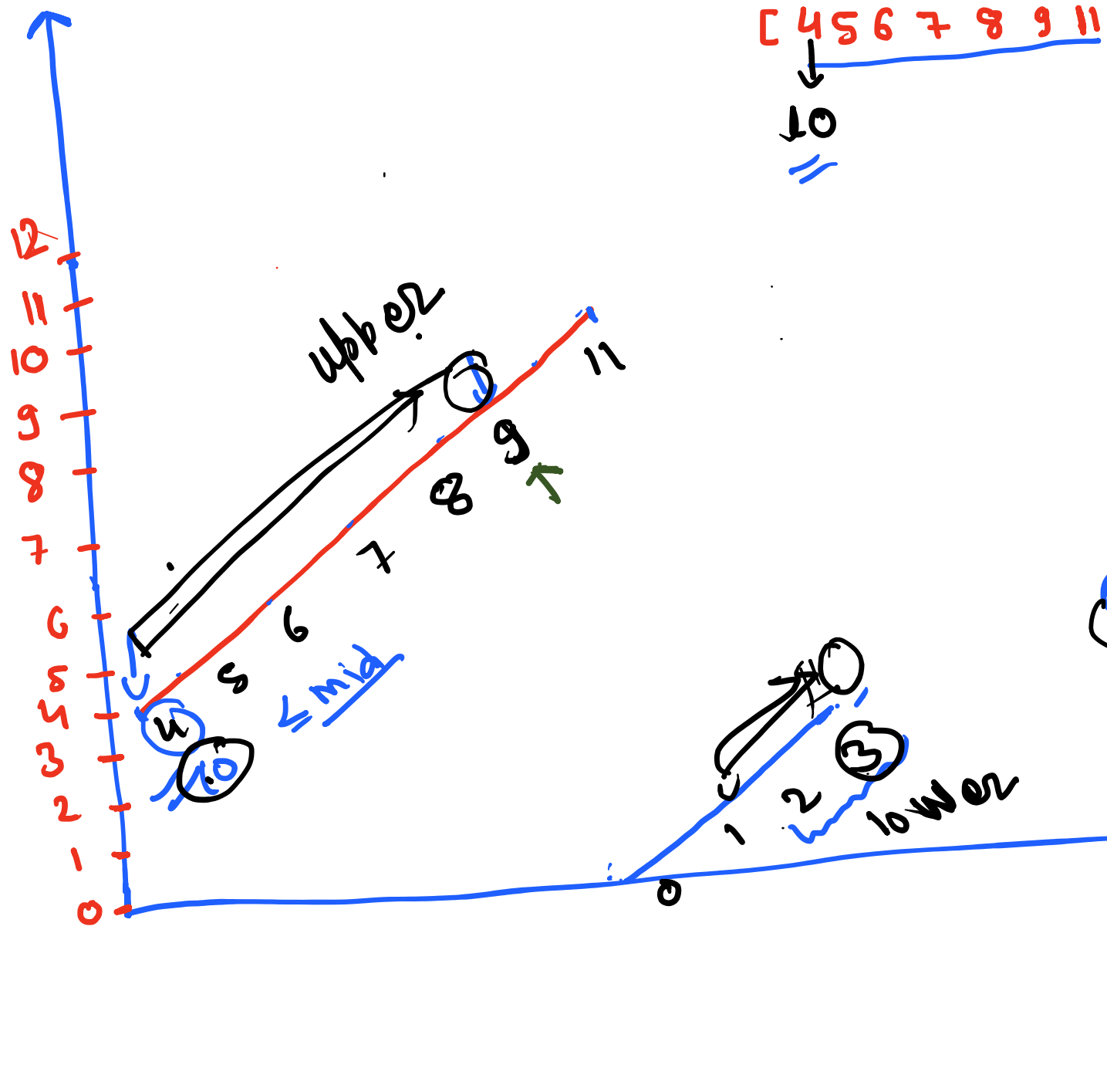
$2^{31}-3$

$$\frac{lo+hi}{2} = \frac{hi-lo}{2} + lo = \frac{hi-lo+2lo}{2}$$

0 1 2 3 4 5 6 7 8 9 10

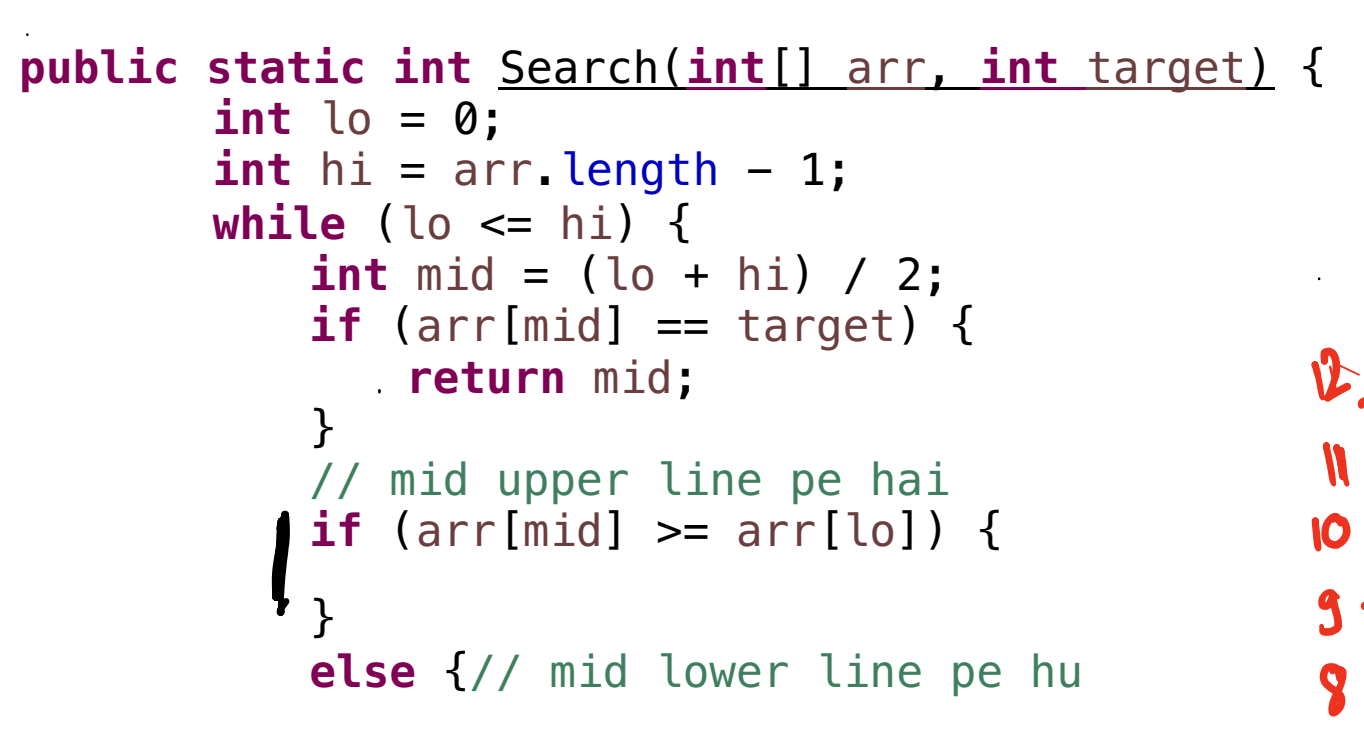
[4 5 6 7 8 9 11 0 1 2 3]

lo hi



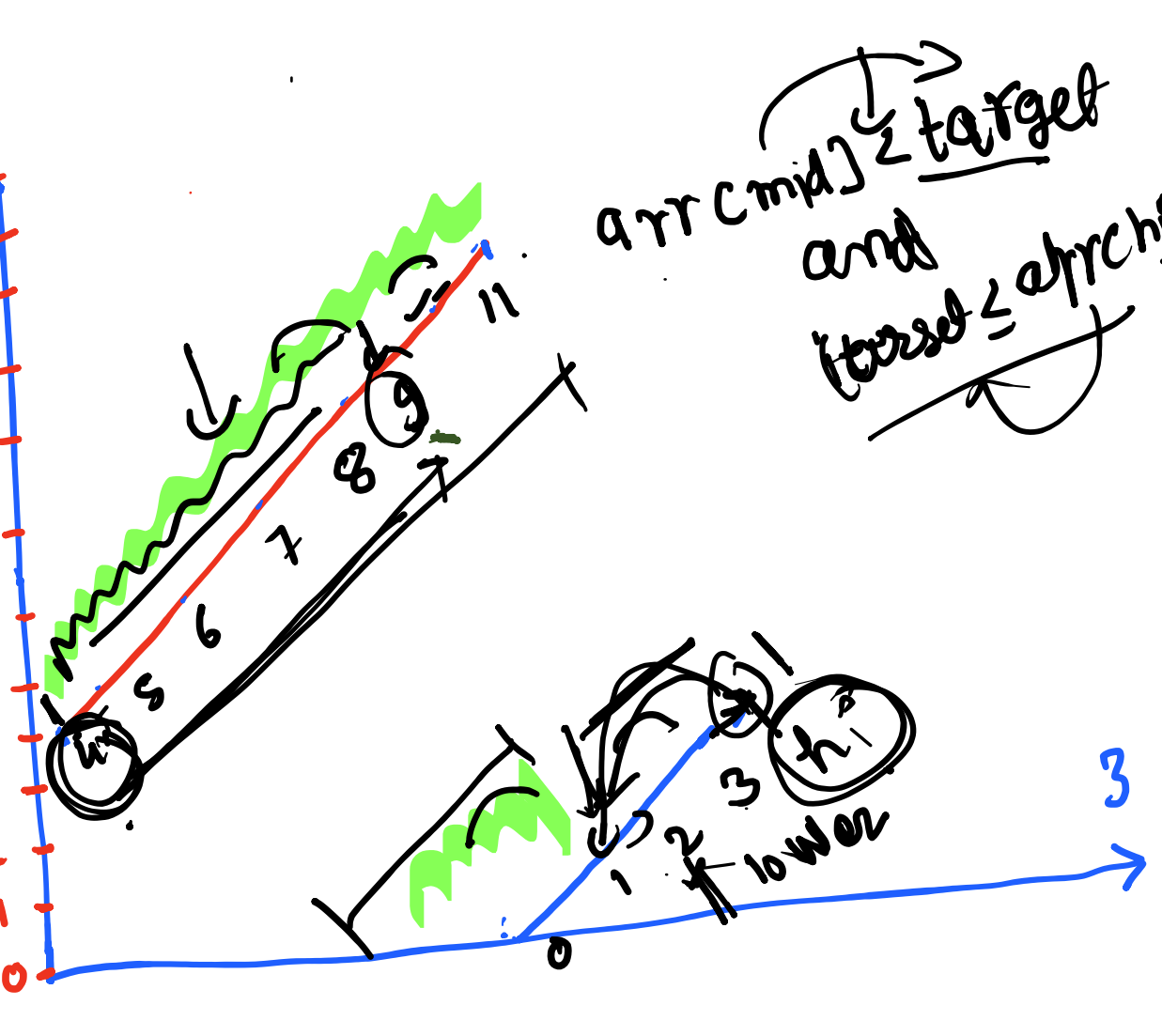
upper
lower
12 11 10 9 8 7 6 5 4 3 2 1 0
4 5 6 7 8 9 11
0 1 2 3
lo
mid

```
while (lo<=hi) {
    mid = (lo+hi)/2
    if (arr[mid] == item) {
        ret mid
    }
    else if (arr[mid] >= arr[lo]) {
    }
    else {
    }
}
```



```
public static int Search(int[] arr, int target) {
        int lo = 0;
        int hi = arr.length - 1;
        while (lo <= hi) {
            int mid = (lo + hi) / 2;
            if (arr[mid] == target) {
                return mid;
            }
            // mid upper line pe hai
            if (arr[mid] >= arr[lo]) {
            }
            else {// mid lower line pe hu
            }
        }
    }
```

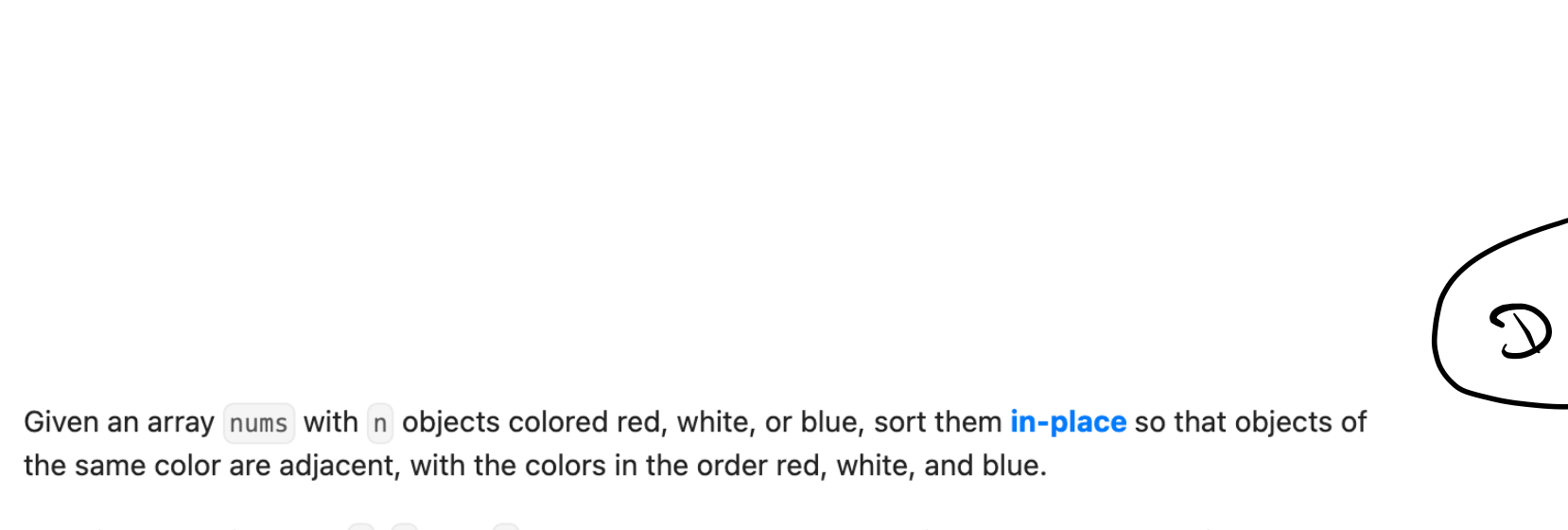
arr[lo] ≤ item && arr[mid] > item



12 11 10 9 8 7 6 5 4 3 2 1 0
4 5 6 7 8 9 11
0 1 2 3
lower
hi

arr[mid] ≥ target and target ≥ arr[lo]

DNF



Given an array `nums` with `n` objects colored red, white, or blue, sort them **in-place** so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

We will use the integers `0`, `1`, and `2` to represent the color red, white, and blue, respectively.

You must solve this problem without using the library's sort function.

Example 1:

```
Input: nums = [2,0,2,1,1,0]
Output: [0,0,1,1,2,2]
```

Example 2:

```
Input: nums = [2,0,1]
Output: [0,1,2]
```

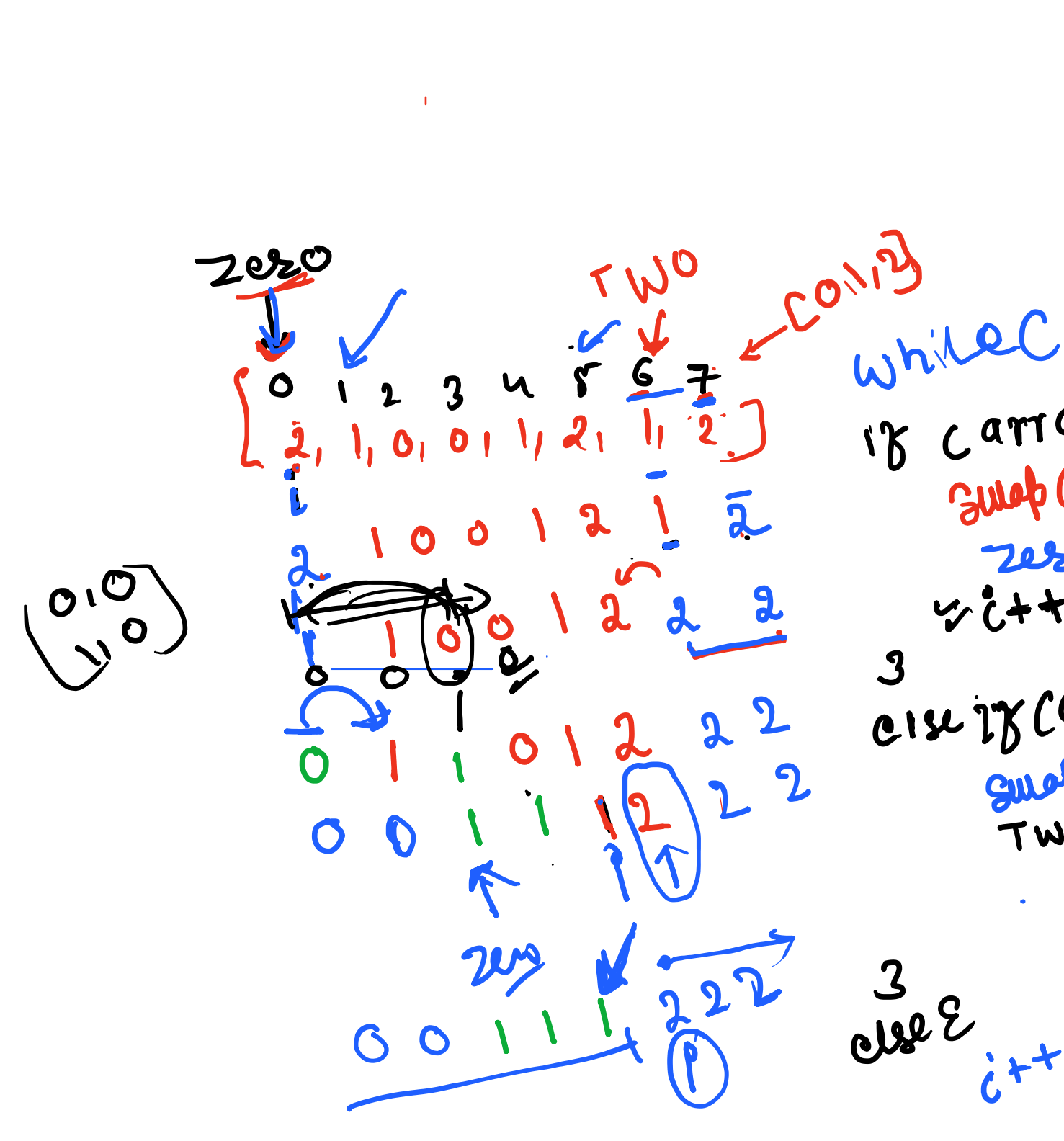
white

Red

Ter

pink

zero=0, Two=arr.len-1



zero
Two
[0,1,2]
0 1 2 3 4 5 6 7
[2, 1, 0, 0, 1, 2, 1, 2]
i
2 1 0 0 1 2 1 2
2 1 0 0 1 2 2 2
0 1 1 0 1 2 2 2
0 0 1 1 1 2 2 2
0 0 1 1 1 2 2 2
zero

```
while (i <= Two) {
    if (arr[i] == 0) {
        swap(i, zero)
        zero++
        i++
    }
    else if (arr[i] == 2) {
        swap(i, Two)
        Two--
    }
    else {
        i++
    }
}
```